

Barren plateaus in quantum neural network training landscapes

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Explanation

1. **Aim:** To prove exponential growth of barren plateaus with increase in number of qubits
2. **Background:** Linear Algebra, Basics of Quantum gates and operations, Parameterized circuits, Variational Quantum Eigen Solvers, Hybrid Quantum-Classical Algorithms
3. **Problem statement:** Evidence of barren plateaus in denser circuits
4. **Methodology:** Authors tried comparing gradient measures of parameterized circuits to Haar Measure to find uniformity
5. **Key Concepts:**
 - a. Haar Measure
 - b. Statistical Moments
 - c. General representation of quantum states and operations
6. **Results:** Authors proved variance of gradients being inversely proportional to 2^{kn} , in each and every case.
7. **Significance:** This shows that even the 1st qubit will have vanishing gradient because of noise from following 2-design circuit
8. **Personal Understanding:**

1. First the paper discusses about parameterized circuits and Haar Measure. They also discuss 2 requirements to prove a space is a barren plateau:
 - a. Mean/Average of gradients – zero
 - b. Variance of gradients – zero
2. Haar measure is kind of a rule that defines uniform unbiased probability distribution.
3. It is valuable since authors express if a parameterized circuit is deep enough twisting parameters (θ) causes the overall operation ($U(\theta)$) to perform as harr measure, hence $\rightarrow \bigcup(\theta) = \bigcup(\theta \pm \theta_k), \bigcup(\theta) = U(\theta) \cdot U(\theta_k), U=U.V$
4. Simulating a Haar Measure on deep circuits is difficult as of now, so authors express that up to 2nd moment of statistics (Variance) a circuit imitates Haar Measure.
5. This results in formation of t-design:
 - a. 1-design: A circuits which follows Haar Measure upto 1st statistical moment (average)
 - b. 2-design: A circuit which follows Haar Measure up to 2nd statistical moment (Variance)
6. Average $\rightarrow E[x]$
7. Variance $\rightarrow E[x^2] - E[x]^2$
8. To measure gradient of a qubit k, circuit is divided into two parts –
9. If either of them follow a 1-design then Average of gardients for qubit k is zero

- a. Then the authors calculate Variance using Levy's lemma and Lipschitz continuity, which results in:

$$\text{Var}[\partial_k E] \approx \begin{cases} -\frac{\text{Tr}(\rho^2)}{(2^{2n}-1)} \text{Tr} \left\langle [V, u^\dagger H u]^2 \right\rangle_{U_+} \\ -\frac{\text{Tr}(H^2)}{(2^{2n}-1)} \text{Tr} \left\langle [V, u \rho u^\dagger]^2 \right\rangle_{U_-} \\ \frac{1}{2^{(3n-1)}} \text{Tr}(H^2) \text{Tr}(\rho^2) \text{Tr}(V^2) \end{cases}$$

10. First case -> U_- is 2-design

Second case -> U_+ is 2-design

Third case -> both are 2-design